

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class :SYBSCIT	Batch : 1
Date of Assignment :22-07-2026	Date/Time of Submission :22-07-2026

Questions:

2a. Write a python code to create nested tuples.

Code:

```
*S55.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/S55.p
File Edit Format Run Options Window Help
student = ("SUNNY",55,("Bsc IT","Semester III"),75)
print("Name:",student[0])
print("Roll:",student[1])
print("Department:",student[2][0])
print("Semester:",student[2][1])
print("Percentage:",student[-1])

print("\nSUNNY SYIT S55")
```

Output:

```
Name: SUNNY
Roll: 55
Department: Bsc IT
Semester: Semester III
Percentage: 75

SUNNY SYIT S55
```

2b. Write a python code to sort the nested tuple using sorted () function.

Code:

```
S55.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/S55.py (3.14
File Edit Format Run Options Window Help
data = [(1, 4),(4, 2),(3, 5)]

sorted_data = sorted(data, key = lambda x: x[1])
print(sorted_data)
print("\nSUNNY SYIT S55")
```

Output:

```
[(4, 2), (1, 4), (3, 5)]
```

```
SUNNY SYIT S55
```

2c. Write a python code to copy or clone list.

Code:

```
S55.py - C:/Users/IT-50/AppData/Local/Programs/Python/Python314/S5
File Edit Format Run Options Window Help
list1 = ["SUNNY", "Amit", "Nitesh"]
list2 = list1.copy()

print("Original List:", list1)
print("Copied List:", list2)

print("\nSUNNY SYIT S55")
```

Output:

```
Original List: ['SUNNY', 'Amit', 'Nitesh']
Copied List: ['SUNNY', 'Amit', 'Nitesh']
```

```
SUNNY SYIT S55
```

```
|
```

2d. Write a python code to check immutability property of python tuples.

Code:

```
S55.py - C:/Users/IT-50/AppData/Local/Programs/Python/Pythor
File Edit Format Run Options Window Help
data = (10, 20, 30)

print("Original Tuple:", data)

data[1] = 50
print(data)

print("\nSUNNY SYIT S55")
```

Output:

```
Original Tuple: (10, 20, 30)
Traceback (most recent call last):
  File "C:/Users/IT-50/AppData/Local/Programs/Python/Python314/S55.py", line 5, in <module>
    data[1] = 50
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80